Generation Is Not Color: Resolving the Symmetric Embedding of Algebraic R-Flux in Gresnigt's Three-Generation Construction

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Abstract

Gresnigt's construction of three Standard Model fermion generations from the complexified sedenions produces three octonion subalgebras O_1, O_2, O_3, related to the physical generation structure by an order-three automorphism ψ. A companion note left open whether Günaydin–Lüst–Malek's octonion-derived algebraic R-flux algebra coincides canonically with one specific O_i, after finding that existence-based isomorphism tests are structurally incapable of answering this, since they hold uniformly for all three regardless of any special relationship between the O_i. We show, by direct computation, that ψ itself does not relate the three subalgebras to one another — it stabilizes each one individually, generating the three fermion generations entirely within a fixed O_i. The symmetry actually responsible for the uniform embedding is a separate, explicitly constructed pair of automorphisms τ_{12}, τ_{13} that together generate an S_3 action permuting O_1, O_2, O_3 exactly. Composing these with a known embedding of GLM's algebra into O_1 reproduces embeddings into O_2 and O_3 found by independent search. Generation and color are orthogonal symmetries of the construction: ψ acts by rotating each of seven index-pairs and never permutes the O_i, while the color symmetry acts by signed permutation of basis indices and never touches the generation ladder. This resolution does not depend on, and does not take a position on, the disputed global isomorphism type of $\mathrm{Aut}(\mathbb{S})$.

1 Background and the question left open

A companion note poses the question of whether $\mathrm{Im}(\mathbb{O})_{\mathrm{GLM}}$, the seven-dimensional Malcev algebra underlying Günaydin, Lüst & Malek's algebraic R-flux construction (arXiv:1607.06474), coincides canonically with one specific member of Gresnigt's three sedenion-derived octonion subalgebras (Gresnigt, *Eur. Phys. J. C* 79, 890, 2019), rather than merely being abstractly isomorphic to it, which Hurwitz's theorem guarantees trivially for any two octonion algebras. That note establishes, by exhaustive computational search, that existence-based tests — does some structure-preserving map exist, subject to natural anchor constraints — cannot distinguish this question from the trivial reading: compatible embeddings were found into all three subalgebras, in large, uniform numbers, regardless of which structurally motivated anchor points were fixed. This was traced to the octonion automorphism group's own rich transitivity, not to any specific relationship between O_1, O_2, O_3 — leaving open whether the uniform embedding reflects a genuine, checkable symmetry of the construction, or is simply an artifact too weak to carry any meaning.

2 An initial hypothesis, tested and found false

Gresnigt's construction carries an explicit order-three automorphism ψ of the full sedenion algebra $\mathbb{S}$, acting on each pair (e_i, e_{i+8}), $i = 1, \ldots, 7$, as a rotation by $2\pi/3$:

$$\psi(e_i) = -\tfrac{1}{2}e_i - \tfrac{\sqrt{3}}{2}e_{i+8}, \qquad \psi(e_{i+8}) = \tfrac{\sqrt{3}}{2}e_i - \tfrac{1}{2}e_{i+8}, \qquad \psi(e_0) = e_0, \quad \psi(e_8) = e_8.$$

This is the automorphism generating Gresnigt's three fermion generations from a single set of ladder operators. Given that three color-labeled octonion subalgebras exist in the construction, it is natural to ask whether ψ is also what relates them to one another.

Lemma 1 (Stabilization)**.** *ψ does not relate O_1, O_2, O_3 to one another; it maps each one entirely into itself.*

Proof. Write $O_j = H_j \oplus H_j e$, the Cayley–Dickson doubling of the quaternion subalgebra $H_j = \mathrm{span}\{e_0\} \cup (\text{Fano line } j)$, where the three Fano lines used are exactly the three lines through the shared point 4 in the standard octonion multiplication table:

$$\{1,4,5\}, \quad \{2,4,6\}, \quad \{3,4,7\}.$$

Concretely,

$$\begin{aligned} O_1 &= \mathrm{span}\{e_0, e_1, e_4, e_5, e_8, e_9, e_{12}, e_{13}\}, \\ O_2 &= \mathrm{span}\{e_0, e_2, e_4, e_6, e_8, e_{10}, e_{12}, e_{14}\}, \\ O_3 &= \mathrm{span}\{e_0, e_3, e_4, e_7, e_8, e_{11}, e_{12}, e_{15}\}. \end{aligned}$$

Each O_j, apart from its two ψ-fixed generators e_0, e_8, consists of exactly three *complete* pairs (e_i, e_{i+8}). Since ψ acts block-diagonally — rotating each pair (e_i, e_{i+8}) within its own two-dimensional span and fixing e_0, e_8 pointwise — it cannot map a generator of O_j outside O_j: every pair that O_j contains is ψ-invariant as a subspace. Hence $\psi(O_j) \subseteq O_j$ for $j = 1, 2, 3$, with no possibility of crossing between them. □

Verification. Confirmed directly on a from-scratch Cayley–Dickson realization of $\mathbb{S}$: $\psi^3 = \mathrm{id}$ and the automorphism identity $\psi(x)\psi(y) = \psi(xy)$ both hold exactly (maximum error $\sim 10^{-16}$, i.e. machine epsilon) across all 16×16 pairs of basis elements; ψ was applied to each generator of O_1, O_2, and O_3 in turn, and every image confirmed to lie entirely within its own subalgebra's span, with exactly zero component elsewhere. Each O_j was independently confirmed to be closed under sedenion multiplication.

3 Confinement of the generations

Lemma 2 (Confinement)**.** *The ladder operators defining Gresnigt's three generations, A_i, B_i, C_i, are built entirely from a single O_i's own generators.*

Verification. For $i = 1$: A_1, B_1, C_1 are linear combinations of exactly $\{e_1, e_5, e_9, e_{13}\} \subset O_1$. The linear dependence $A_1 + aB_1 + bC_1 = 0$ stated in Gresnigt's construction was confirmed to machine precision for $i = 1, 2, 3$, using the explicit ψ-image coefficients $a = (\sqrt{3} - 1)/2$, $b = (-\sqrt{3} - 1)/2$.

Together, Lemmas 1 and 2 establish that the generation index (which of A, B, C) and the color index (which of O_1, O_2, O_3) are structurally orthogonal in Gresnigt's own construction — not two names for the same threefold symmetry.

4 The symmetry relating O_1, O_2, O_3: an explicit construction

Since ψ has been shown to stabilize each O_i (Lemma 1), any symmetry relating O_1, O_2, O_3 to one another must be a genuinely different automorphism. Rather than locate it inside an asserted global decomposition of $\mathrm{Aut}(\mathbb{S})$, we construct it directly and verify it directly.

Lemma 3 (Diagonal extension)**.** *Let $\varphi \in \mathrm{Aut}(\mathbb{O})$ be any automorphism of the octonions. Then the map $\Phi(a,b) := (\varphi(a), \varphi(b))$ is an automorphism of $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}$.*

Proof. Every automorphism of a composition algebra preserves the norm form and hence commutes with conjugation: $\varphi(\bar{x}) = \overline{\varphi(x)}$. Using the Cayley–Dickson product $(a,b)(c,d) = (ac - \bar{d}b,\ da + b\bar{c})$ and linearity of φ,

$$\Phi\big((a,b)(c,d)\big) = \big(\varphi(ac) - \varphi(\bar{d}b),\ \varphi(da) + \varphi(b\bar{c})\big) = \big(\varphi(a)\varphi(c) - \overline{\varphi(d)}\varphi(b),\ \varphi(d)\varphi(a) + \varphi(b)\overline{\varphi(c)}\big),$$

using that φ is a homomorphism on $\mathbb{O}$. This is exactly $\big(\varphi(a), \varphi(b)\big)\big(\varphi(c), \varphi(d)\big) = \Phi(a,b)\,\Phi(c,d)$. □

This is a general fact about Cayley–Dickson doubling, independent of any claim about $\mathrm{Aut}(\mathbb{S})$'s global structure: *any* automorphism of the 8-dimensional block extends diagonally to an automorphism of the 16-dimensional algebra, automatically.

Construction. The (independently verified, order-1344) signed-permutation automorphism group of the octonion sub-block $e_1, \ldots, e_7$ contains 64 elements sending the pair $\{1,5\}$ to the pair $\{2,6\}$ while fixing 4, and 64 sending $\{1,5\}$ to $\{3,7\}$ while fixing 4. Among each set of 64, an order-2 representative was located; call their diagonal extensions (Lemma 3) τ_{12} and τ_{13}.

Lemma 4 (Color automorphisms)**.** *There exist $\tau_{12}, \tau_{13} \in \mathrm{Aut}(\mathbb{S})$, each of order 2, with*

$$\tau_{12}(O_1) = O_2, \quad \tau_{12}(O_2) = O_1, \quad \tau_{12}(O_3) = O_3,$$

$$\tau_{13}(O_1) = O_3, \quad \tau_{13}(O_3) = O_1, \quad \tau_{13}(O_2) = O_2.$$

Consequently $\tau_{12} \circ \tau_{13}$ has order 3 and cyclically permutes $O_1 \to O_3 \to O_2 \to O_1$. Two distinct transpositions generate the full symmetric group on three letters, so $\langle \tau_{12}, \tau_{13} \rangle \cong S_3$ acts on $\{O_1, O_2, O_3\}$ as the complete permutation group.

Verification. Both τ_{12} and τ_{13} were built exactly as in the Construction and checked to be genuine automorphisms of the full 16-dimensional sedenion algebra directly — i.e. against all ordered pairs of sedenion basis products, not merely inferred from Lemma 3 — with zero error. $\tau_{12}^2 = \mathrm{id}$ and $\tau_{13}^2 = \mathrm{id}$ were confirmed exactly. The action on O_1, O_2, O_3 stated above was confirmed by applying each map to every generator of each subalgebra and checking membership in the claimed target's span; $\tau_{12} \circ \tau_{13}$ was confirmed to have order exactly 3 and to send $O_1 \to O_3$.

Remark. This is deliberately a claim about a specific, exhibited subgroup of $\mathrm{Aut}(\mathbb{S})$, not about $\mathrm{Aut}(\mathbb{S})$ as a whole. It does not require, and does not assume, any particular answer to whether $\mathrm{Aut}(\mathbb{S}) \cong G_2$ or $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$ globally — a question this note is not equipped to adjudicate and on which the cited literature itself disagrees. What Lemma 4 shows is narrower and fully self-contained: two explicitly constructed, directly verified automorphisms exist, they are of a visibly different structural kind from ψ (signed permutation of basis indices, versus ψ's block-diagonal rotation of index pairs), and together they do exactly what is needed — generating an S_3 that permutes O_1, O_2, O_3 while touching none of the generation structure confined by Lemma 2.

5 The composition theorem

Theorem 1. *Let $f : \mathrm{Im}(\mathbb{O})_{\mathrm{GLM}} \to O_1$ be an embedding found by the direct search of the companion note. Then $\tau_{12} \circ f$ is exactly one of the embeddings found by an independent direct search into O_2; likewise $\tau_{13} \circ f$ exactly reproduces an independently-found embedding into O_3.*

This is the substantive result: the uniform existence of embeddings into all three O_i, previously shown to be too weak a fact to carry meaning on its own, is here explained by a specific, verified, non-trivial automorphism of the ambient sedenion structure — not merely accompanied by one.

Remark. One precise gap worth flagging for reproducibility: the Construction above shows 64 valid choices exist for an automorphism with the defining property used to build τ_{12} (and likewise 64 for τ_{13}). The Composition Theorem's exact match requires a *specific* choice, not merely membership in this set of 64. Which representative was used should be recorded explicitly rather than left implicit, so that the exact-match claim is independently checkable by a third party.

6 What this resolves

The original question — does GLM's algebra coincide canonically with one specific O_i — is answered here, precisely, in the negative, and for a stronger reason than mere existence-symmetry. It is not merely that embeddings happen to exist into all three; it is that a genuine, checked symmetry of the full construction carries any one embedding onto the others exactly. The three embeddings are not independent facts each separately requiring explanation — they are the same fact, related by the explicitly constructed S_3 of Lemma 4. There is no privileged O_i for algebraic R-flux to canonically prefer, and this is established rather than merely undecidable.

It is worth being precise about what this does not show: it does not show that GLM's construction is related to Gresnigt's own generation structure in any physically meaningful way — ψ, the symmetry actually tied to the physics of three fermion generations, plays no role in this result at all (Lemma 1). The resolution concerns the algebra's internal color symmetry, not a statement about which fermion generation, if any, algebraic R-flux might be associated with.

7 Scope

Scope. The explicit form of ψ; $\psi^3 = \mathrm{id}$ and its automorphism property; Lemma 1 (stabilization), both by direct computation and by the structural block-diagonal argument given above; closure of O_1, O_2, O_3 under sedenion multiplication; the 1344-element signed-permutation automorphism group of the octonion sub-block; and the existence, automorphism property, order, and action of τ_{12} and τ_{13} (Lemmas 3–4) all rest on a from-scratch Cayley–Dickson construction of $\mathbb{O}$ and $\mathbb{S}$, checked numerically to machine precision throughout.

Lemma 2's ladder-operator coefficients follow Gresnigt's own construction directly; confirming them against her published formulas, rather than via the linear-dependence check reported here, has not been done. The Composition Theorem's exact match requires the companion note's independently-found embedding lists into O_2 and O_3; it cannot be reproduced from this note alone without that note's search output. Whether every embedding into O_i composes correctly under $\psi, \tau_{12}, \tau_{13}$ — rather than only the specific representatives used throughout — has not been checked.

References

1. Gresnigt, N. G., "Three fermion generations with two unbroken gauge symmetries from the complex sedenions," *Eur. Phys. J. C* 79, 890 (2019).

2. Gresnigt, N. G., Gourlay, J. & Varma, K., "Three generations of colored fermions with family symmetry from Cayley-Dickson sedenions," arXiv:2306.13098.

3. Günaydin, M., Lüst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes," arXiv:1607.06474.
4. Wilmot, G. P., "Automorphisms of Sedenions," arXiv:2512.07210 (2025). The paper's own conclusion is $\mathrm{Aut}(\mathbb{S}) \cong G_2$, in agreement with Schafer and against a $G_2 \times S_3$ claim it attributes to Brown; it is cited here only for that context, not as support for any claim used in this note.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.